

Кладковой Максим 5130904/30002 Вар 47
 Кусочно-кубическая интерполяция Эрмита

$$f(x) = \sin(\pi x) \cdot x \in [-\frac{1}{4}; \frac{1}{4}] ; n = 3 ; \frac{df}{dx} = \pi \cos(\pi x)$$

No	X	f(x)	$\frac{df}{dx} = m$
0	$-\frac{1}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2}$
1	$-\frac{1}{6}$	$-\frac{1}{2}$	$\frac{\pi\sqrt{3}}{2}$
2	0	0	π
3	$\frac{1}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2}$

$$h_1 = -\frac{1}{6} + \frac{1}{4} = \frac{1}{12} \quad t = \frac{x + \frac{1}{4}}{\frac{1}{12}} = 12x + 3 \quad x \in [-\frac{1}{4}; -\frac{1}{6}]$$

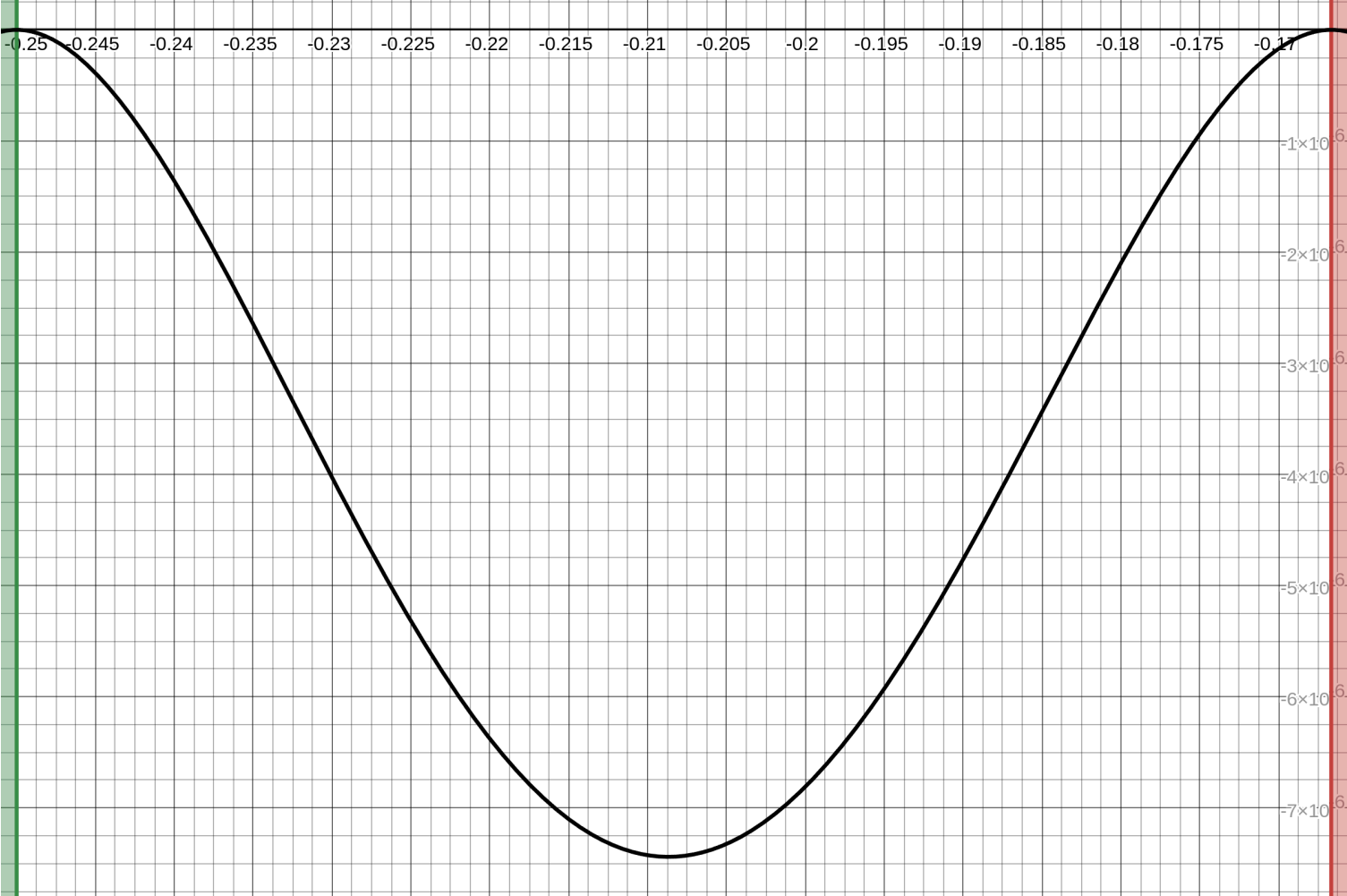
$$H_1(t) = -\frac{\sqrt{2}}{2}(1 - 3t^2 + 2t^3) - \frac{1}{2}(3t^2 - 2t^3) + \frac{\pi\sqrt{2}}{2} \frac{1}{12}(t - 2t^2 + t^3) + \frac{\pi\sqrt{3}}{2} \frac{1}{12}(-t^2 + t^3)$$

$$\begin{aligned} H_1(x) &= -\frac{\sqrt{2}}{2}(1 - 3(12x+3)^2 + 2(12x+3)^3) - \frac{1}{2}(3(12x+3)^2 - 2(12x+3)^3) + \frac{\pi\sqrt{2}}{24}(12x+3 - 2(12x+3)^2 + (12x+3)^3) + \\ &+ \frac{\pi\sqrt{3}}{24}(-12x+3)^2(12x+3)^3 = -\frac{(1728x^3 + 1080x^2 + 216x + 14)\sqrt{2}}{2} + \frac{3456x^3 + 2160x^2 + 432x + 27}{2} + \frac{\pi(16x^4 + 144x^3 + 144x^2)}{4} + \frac{\pi\sqrt{3}(192x^3 + 42x + 3 + 288)}{4} \\ &= x^3(-1728\sqrt{2} + 1728 + 72\pi\sqrt{2} + 72\pi\sqrt{3}) + x^2(-1080\sqrt{2} + 1080 + 42\pi\sqrt{2} + 48\pi\sqrt{3}) + \\ &+ x(-216\sqrt{2} + 216 + 8\pi\sqrt{2} + \frac{21\pi\sqrt{3}}{2}) + (-14\sqrt{2} + \frac{27}{2} + \frac{\pi\sqrt{2}}{2} + \frac{3\pi\sqrt{3}}{4}) \end{aligned}$$

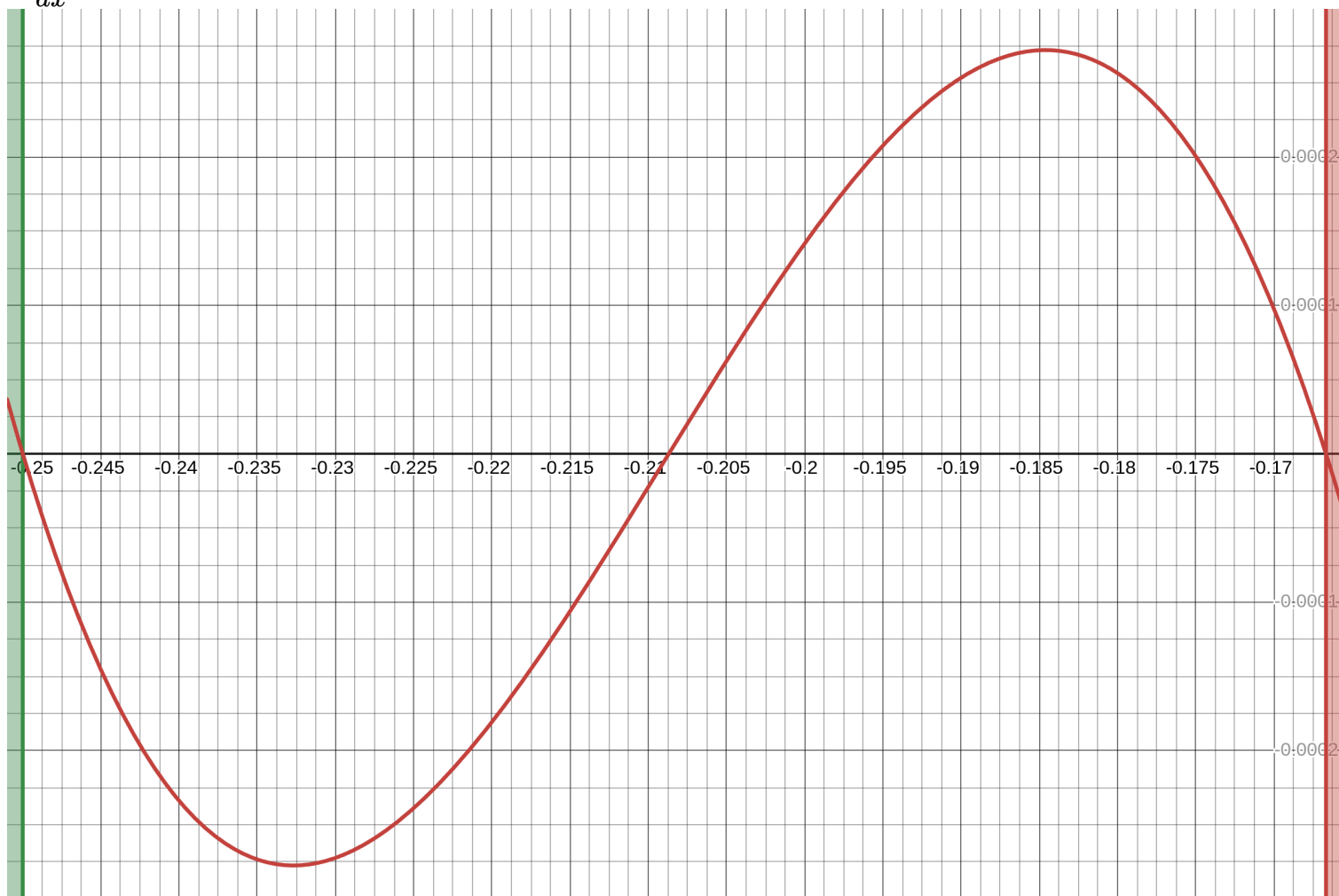
$$\begin{aligned} \Gamma_1(x) &= \sin(\pi x) - x^3(-1728\sqrt{2} + 1728 + 72\pi\sqrt{2} + 72\pi\sqrt{3}) + x^2(-1080\sqrt{2} + 1080 + 42\pi\sqrt{2} + 48\pi\sqrt{3}) + \\ &+ x(-216\sqrt{2} + 216 + 8\pi\sqrt{2} + \frac{21\pi\sqrt{3}}{2}) + (-14\sqrt{2} + \frac{27}{2} + \frac{\pi\sqrt{2}}{2} + \frac{3\pi\sqrt{3}}{4}) \end{aligned}$$

$$\begin{aligned} \frac{d\Gamma_1(x)}{dx} &= \pi \cos(\pi x) - 3x^2(-1728\sqrt{2} + 1728 + 72\pi\sqrt{2} + 72\pi\sqrt{3}) + 2x(-1080\sqrt{2} + 1080 + 42\pi\sqrt{2} + 48\pi\sqrt{3}) + \\ &+ (-216\sqrt{2} + 216 + 8\pi\sqrt{2} + \frac{21\pi\sqrt{3}}{2}) + (-14\sqrt{2} + \frac{27}{2} + \frac{\pi\sqrt{2}}{2} + \frac{3\pi\sqrt{3}}{4}) \end{aligned}$$

$r_1(x)$:



$$\frac{dr_1(x)}{dx}:$$



$$h_2 = 0 + \frac{1}{6} = \frac{1}{6}; t_2 = \frac{x + \frac{1}{6}}{\frac{1}{6}} = 6x + 1 \quad x \in \left[-\frac{1}{6}, 0\right]$$

$$H_2(t) = -\frac{1}{2}(1 - 3t^2 + 2t^3) + 0(3t^2 - 2t^3) + \frac{\pi\sqrt{3}}{2}\frac{1}{6}(1 - 2t^2 + t^3) + \frac{\pi}{6}(-t^2 + t^3) =$$

$$= -\frac{(1 - 3t^2 + 2t^3)}{2} + \frac{\pi\sqrt{3}(1 - 2t^2 + t^3)}{12} + \frac{\pi(-t^2 + t^3)}{6}$$

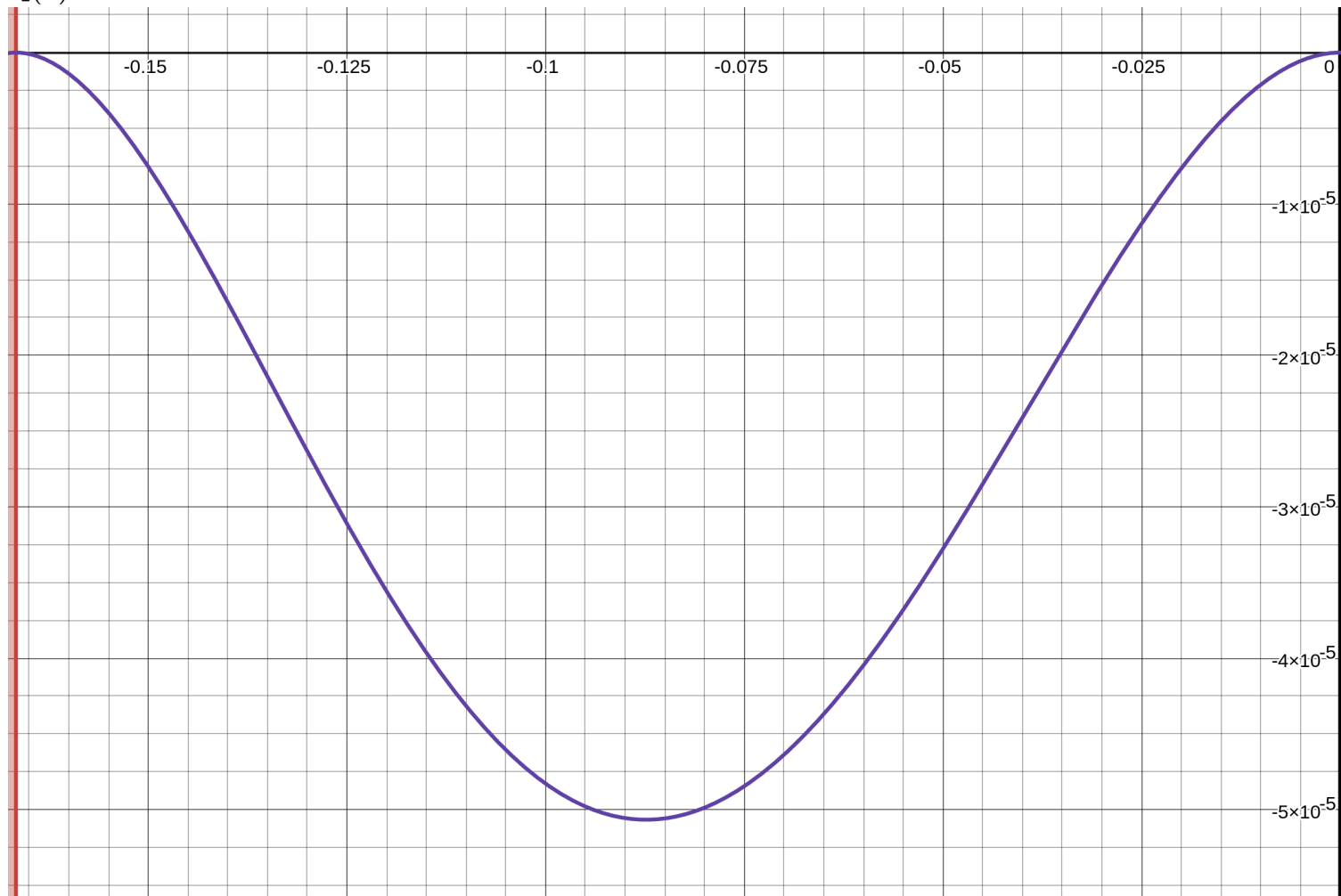
$$H_2(x) = \frac{-(1 - 3(6x+1)^2 + 2(6x+1)^3)}{2} + \frac{\pi\sqrt{3}(6x+1 - 2(6x+1)^2 + (6x+1)^3)}{12} + \frac{\pi(-(6x+1)^2 + (6x+1)^3)}{6} =$$

$$= x^3(-216 + 18\pi\sqrt{3} + 36\pi) + x^2(-54 + 3\pi\sqrt{3} + 12\pi) + x\pi$$

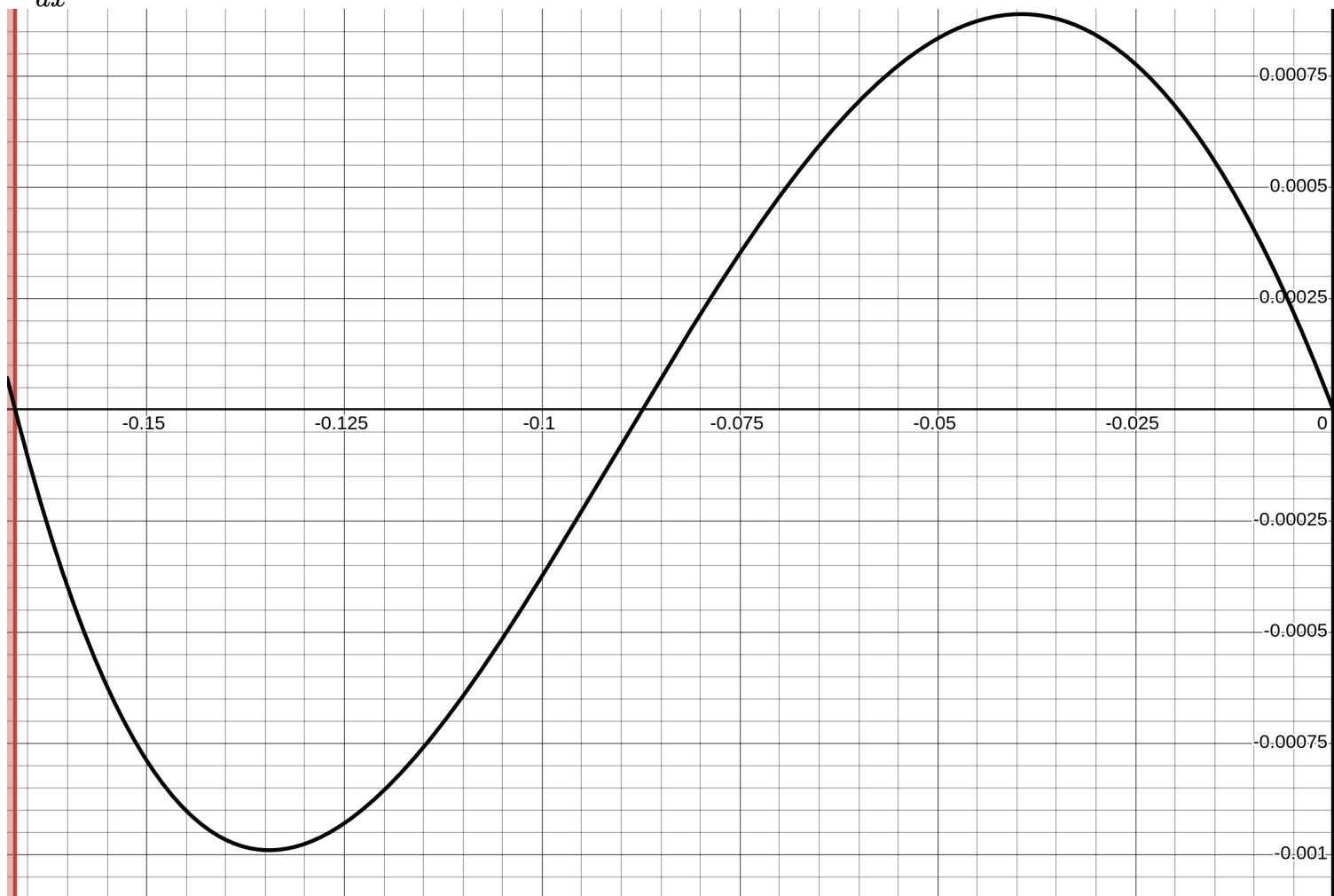
$$r_2(x) = \sin(\pi x) - x^3(-216 + 18\pi\sqrt{3} + 36\pi) + x^2(-54 + 3\pi\sqrt{3} + 12\pi) + x\pi$$

$$\frac{dr_2(x)}{dx} = x \cos(\pi x) - 3x^2(-216 + 18\pi\sqrt{3} + 36\pi) + 2x(-54 + 3\pi\sqrt{3} + 12\pi) + \pi$$

$r_2(x)$:



$$\frac{dr_2(x)}{dx}:$$



$$h_3 = \frac{1}{4} - 0 = \frac{1}{4}; t_3 = \frac{x-0}{\frac{1}{4}} = 4x \quad x \in [0; \frac{1}{4}]$$

$$H_3(t) = 0(1 - 3t^2 + 2t^3) + \frac{\sqrt{2}}{2}(3t^2 - 2t^3) + \frac{\pi}{4}(t - 2t^2 + t^3) + \frac{\pi\sqrt{2}}{2} \frac{1}{4}(-t^2 + t^3) =$$

$$= \frac{3t^2 - 2t^3}{\sqrt{2}} + \frac{\pi(t - 2t^2 + t^3)}{4} + \frac{\pi\sqrt{2}(-t^2 + t^3)}{4\sqrt{2}}$$

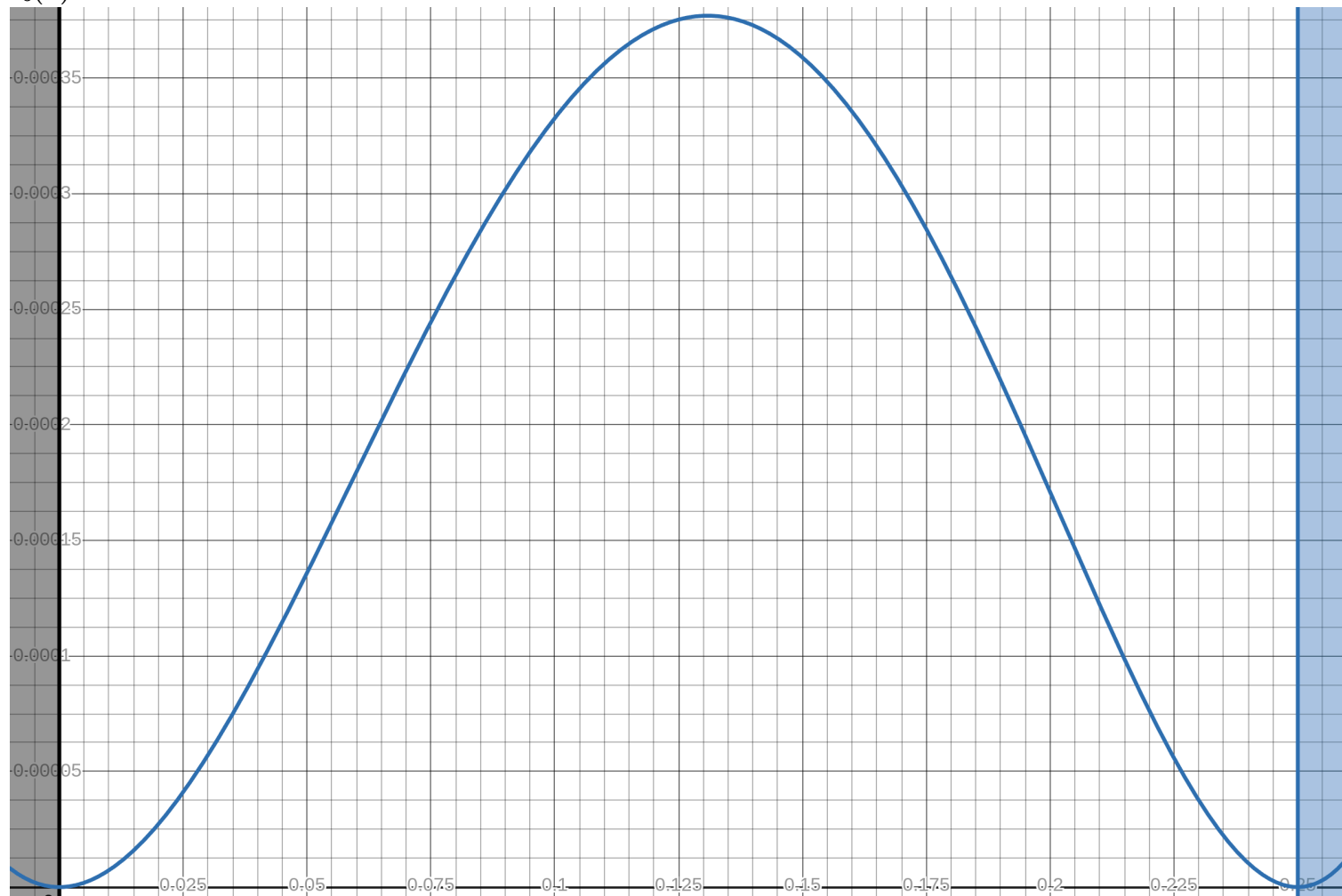
$$H_3(x) = \frac{3(4x)^2 - 2(4x)^3}{\sqrt{2}} + \frac{\pi(4x - 2(4x)^2 + (4x)^3)}{4} + \frac{\pi(-(4x)^2 + (4x)^3)}{4\sqrt{2}} = x^3(-64\sqrt{2} + 16\pi + 8\pi\sqrt{2}) +$$

$$x^2(24\sqrt{2} - 8\pi - 2\pi\sqrt{2}) + \pi x$$

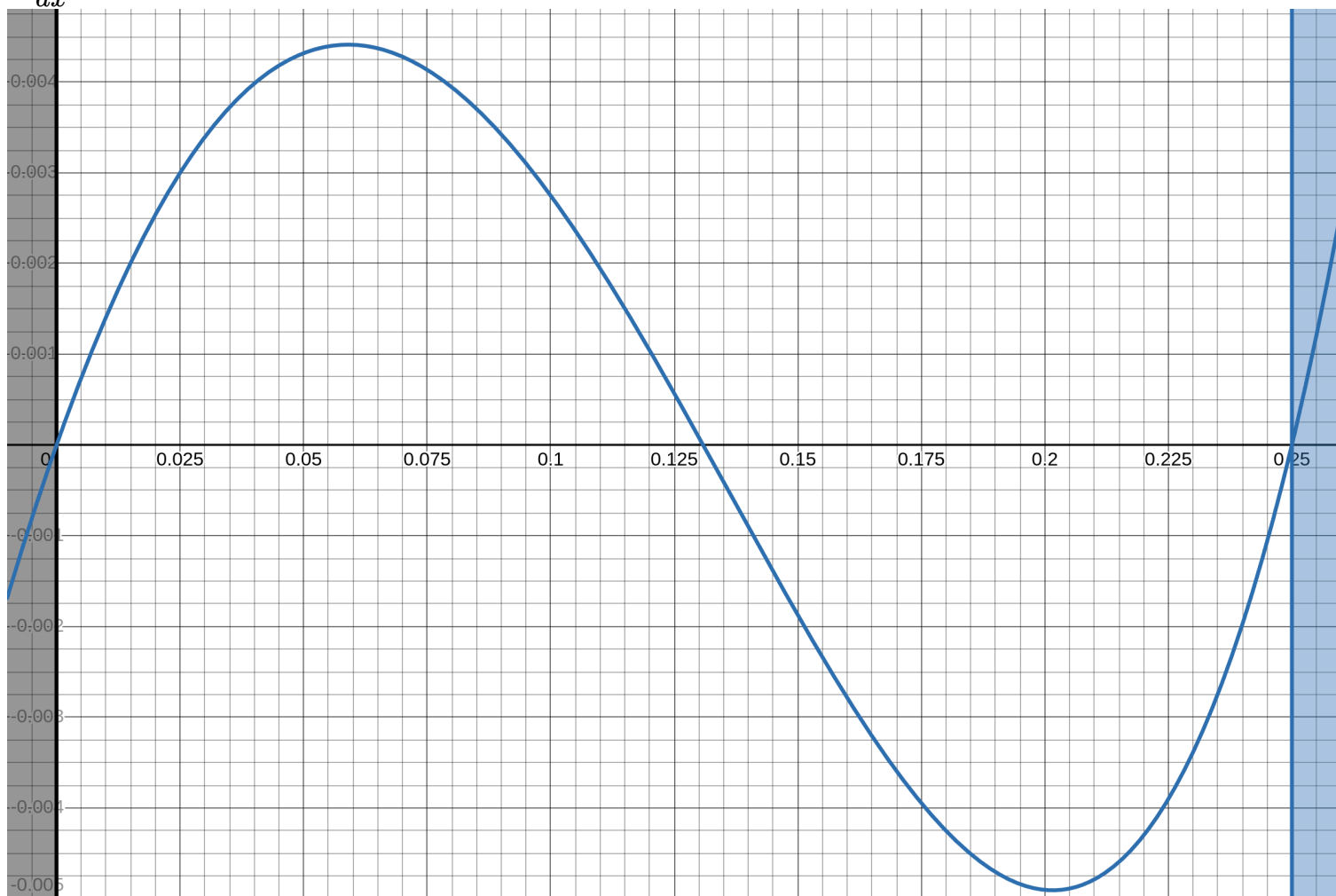
$$f_3(x) = \sin(\pi x) - x^3(-64\sqrt{2} + 16\pi + 8\pi\sqrt{2}) - x^2(24\sqrt{2} - 8\pi - 2\pi\sqrt{2}) - \pi x$$

$$\frac{df_3(x)}{dx} = \pi \cos(\pi x) - 3x^2(-64\sqrt{2} + 16\pi + 8\pi\sqrt{2}) - 2x(24\sqrt{2} - 8\pi - 2\pi\sqrt{2}) - \pi$$

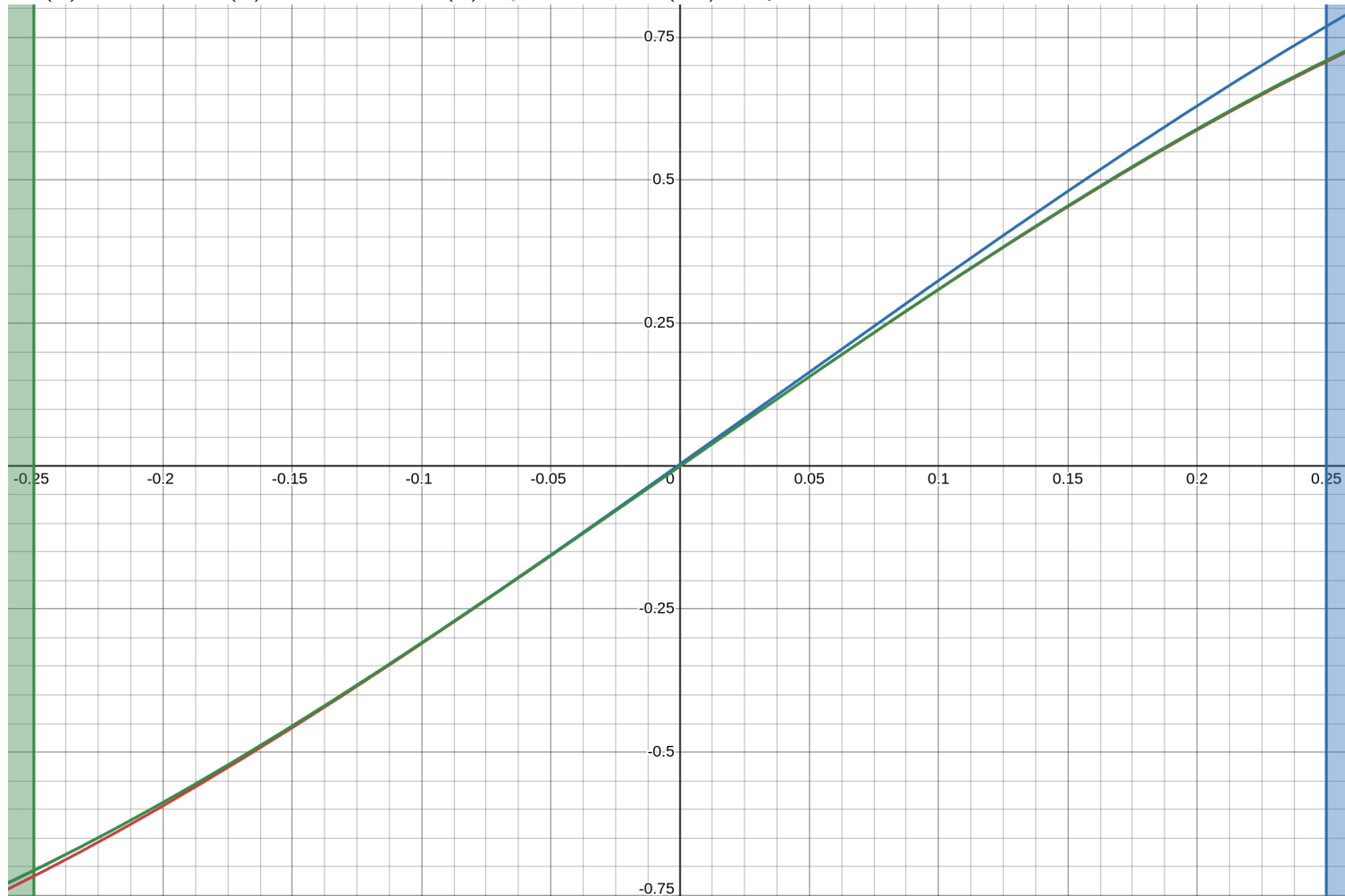
$r_3(x)$:



$$\frac{dr_3(x)}{dx}:$$



$H_1(x)$ -Синий, $H_2(x)$ -Зеленый, $H_3(x)$ -Красный, $\sin(\pi x)$ -Черный



$H_1(x)$ -Красный, $H_2(x)$ -Зеленый, $H_3(x)$ -Желтый, $\sin(\pi x)$ -Синий

$[\sin(3.141593 \cdot x), -4.085 \cdot x^3 + 0.437 \cdot x^2 + 3.2085 \cdot x + 0.00358, -4.958 \cdot x^3 + 0.0233 \cdot x^2 + 3.141593 \cdot x, (-0.0774$

